



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS: X
SUBJECT: CHEMISTRY

FULL MARKS: 40
DATE: 17.10.2022

General Instructions:

- The paper consists of three printed pages.
- Answers should be to the point.
- Question numbers should be copied carefully while answering the questions.
- Marks will be deducted for spelling errors.

Question:1

Select the correct option:

[5]

- i) Two consecutive members of a homologous series differ in their formula by:
- A) CH_2
 - B) CH_3
 - C) C_2H_2
 - D) C_3H_3
- ii) The difference in molecular mass between butyne and pentyne is:
- A) 13
 - B) 14
 - C) 12
 - D) 11
- iii) The metal which is not present in stainless steel is:
- A) Aluminium
 - B) Iron
 - C) Chromium
 - D) Manganese
- iv) The molecular mass of octane is 114, its vapour density is:
- A) 56
 - B) 57
 - C) 55
 - D) 54
- v) Which of the following is not a sulphide ore?
- A) Haematite
 - B) Galena
 - C) Cinnabar
 - D) Zinc blende

Question:2

Match the alloys given in column 1 with the uses given in column 2:

[5]

Column 1	Column 2
i) Duralumin	A) Electrical fuse
ii) Solder	B) Surgical instruments
iii) Brass	C) Aircraft body
iv) Bronze	D) Decorative articles
v) Stainless steel	E) Statues and medals

Question: 3

A gaseous hydrocarbon contains 82.76% of carbon, given that its vapour density is 29 find its molecular formula. [C = 12, H = 1]

[5]

Question: 4

Write the structural formula of the following:

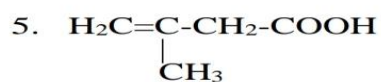
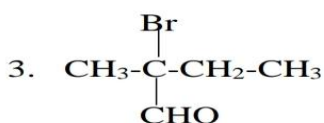
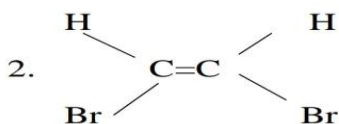
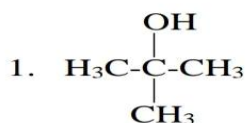
[5]

- i) Methanal
- ii) Propan-2-ol
- iii) Acetone
- iv) Ethanoic acid
- v) 1, 2-Dichloroethane.

Question: 5

Write the IUPAC name of the following:

[5]



Question: 6

- i) 1250 cc of oxygen was burnt with 300 cc of ethane. Calculate the volume of unused oxygen and the volume of carbon dioxide formed. The equation for the reaction is: $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$ [3]
- ii) 1000 ml of a gas X weighs 1 g, calculate the vapour density and molecular mass of the gas if 1 litre of hydrogen weighs 0.09 g at STP. [2]

Question: 7

With respect to extraction of aluminium answer the following: [10]

- i) Name the chief ore of aluminium and also write its formula.
- ii) Why are the anodes replaced periodically during electrolysis of fused alumina. Name one main impurity associated with the ore.
- iii) Write three balanced equations for the reaction taking place during the concentration of the ore.
- iv) Explain why aluminium cannot be obtained by reducing its oxide using reducing agents.
- v) Mention the role of the following in Hall Heroult's process:
 - a) Graphite
 - b) Powdered coke